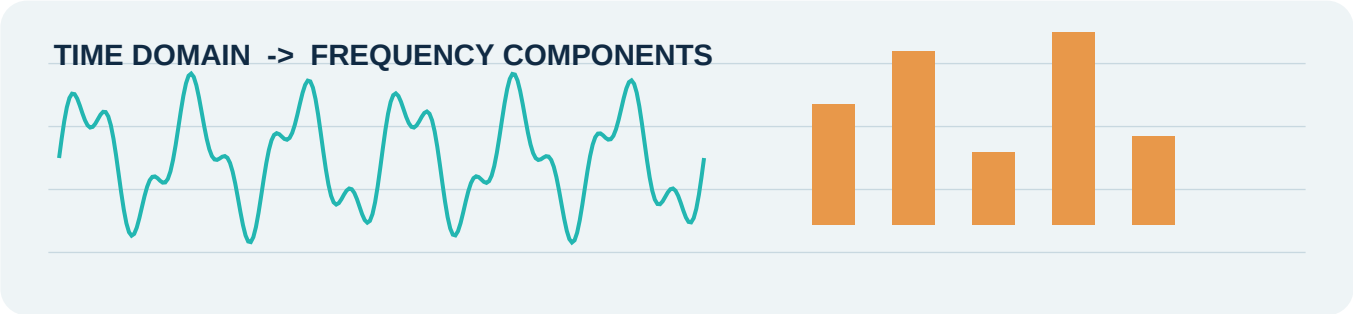
Fourier analysis, sampling and spectra

How sampled signals become mathematical evidence

This original Atlas teaching brief introduces scientific foundations and links to primary literature. It is not a third-party research paper, peer-reviewed original result, or replacement for source documents.



A signal is not its spectrum

A time-domain trace records variation with time. A Fourier representation describes component frequencies and phases. Neither representation alone establishes a cause.

Sampling theorem

A properly bandlimited signal of bandwidth B can be reconstructed under the classical sampling assumptions from sufficiently rapid uniform samples. Real analog filters, finite records, timing drift and noise create departures from ideal assumptions.

Aliasing thought experiment

A 7 Hz sinusoid sampled at 10 samples/second is indistinguishable at those sample instants from an appropriately signed 3 Hz sinusoid. At a different sample rate those sample values can diverge.

A time-frequency tradeoff

Short Fourier analysis windows localize rapid changes but blur nearby frequencies. Long windows better separate tones while smearing brief events. Adaptive windows are studied in PUB-002; reported new improvements should be checked against reproducible baselines.

Resolution is not just bit depth

Timebase stability, window function, analog bandwidth, quantization and SNR affect estimates. Include units, data lineage and confidence intervals for every graph.

Sources and further reading

PUB-002: Universal STFT (arXiv:2610.37795, CC BY 4.0); PUB-003: Survey on sampling recovery (arXiv:2601.08787, CC BY 4.0); PUB-012: MIT OCW Signals and Systems (CC BY-NC-SA terms).